

Sakshyam Sigdel

+1 (601) 913-0628 | sakshyamsigdel@usm.edu | [LinkedIn](#) | [GitHub](#) | [Website](#) | Hattiesburg, MS

PROJECTS

Curtain | *Python, Redis, PostgreSQL, Docker, Prometheus* MLH Prod Eng. Hackathon – Winner

- Won the Scalability Track by architecting a load-balanced web service with a resilient PostgreSQL fallback.
- Optimized heavy data ingestion by implementing Redis sharding and asynchronous stream consumers.
- Integrated end-to-end observability and automated alerting using Prometheus, Grafana, and Discord relays.

SenseNav | *Arduino, Ultrasound, MiDaS, Suno AI* HackMIT – Sept. 2025

- Built an assistive navigation prototype for visually impaired users combining haptic feedback and spatialized audio.
- Integrated ultrasound sensors with monocular depth estimation to reconstruct spatial soundscapes.
- Selected as a top 12 project out of 320+ teams for innovative hardware-software integration.

Acroswe | *Python, Pandas, Ethereum, Solidity* HackPrinceton (MLH) – Spring 2025 Winner

- Won "Best Use of AI and Blockchain" for building a heuristic-based ML model to estimate dynamic property values.
- Implemented smart contracts on the Ethereum blockchain to facilitate transparent and secure asset transfers.

HyprDrover | *Rust, Hyprland IPC, GitHub Actions, AUR* Personal Project – 2025

- Engineered a Rust CLI session manager to snapshot and restore Wayland window layouts via IPC and JSON.
- Automated Arch User Repository (AUR) distribution and configured GitHub Actions CI/CD for publishing.

EXPERIENCE

Undergraduate Researcher — Agentic AI Systems Jan. 2026 – Present
University of Southern Mississippi, PI: Dr. Rabab Abdelfattah Hattiesburg, MS

- Developing a Python-based agent orchestration layer to coordinate vision models (smoke/fire detection) with planning agents for autonomous UAV flight decisions.
- Designed a replayable logging pipeline with per-agent trace IDs to monitor and debug non-deterministic multi-agent behavior and tool-use reliability.

Math Learning Assistant (LA) Aug. 2025 – Dec. 2025
University of Southern Mississippi Hattiesburg, MS

- Tutored students in college-level mathematics, facilitating active learning during lectures.
- Collaborated with faculty to grade assignments and lead supplemental instruction sessions.

TECHNICAL & SOFT SKILLS

Languages: Python, C/C++, Rust, JavaScript, Bash, SQL

Frameworks: React, Node.js, Flask, FastAPI, Pandas, NumPy

Tools & DevOps: Linux, Git, GitHub Actions, Docker, Nginx, DigitalOcean, GDB, strace, tcpdump

Hardware & EE: KiCad, Arduino, MATLAB/Simulink

EDUCATION

University of Southern Mississippi Hattiesburg, MS
Bachelor of Science in Computer Engineering Expected Graduation: 2028

HONORS

MLH Production Engineering Hackathon — Winner Apr. 2026
Winner in the Scalability Track with production-ready infrastructure (Curtain).

USM President's List — 4.0 GPA Spring 2024
Academic excellence award for maintaining a perfect GPA.

Voxo Hatchathon Winner (County) Nov. 2025
Awarded \$1,000 Prize for technical and entrepreneurial excellence. (Univ. of Southern Mississippi, MS)

Hack@Davidson (MLH) Winner Feb. 2025
2nd Overall at Major League Hacking event (Davidson College, NC)